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IntelliDine: Machine Learning Driven Queue and Service Management System

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***Abstract*— Campus canteens often face severe congestion during peak hours, resulting in long queues, inefficient service, and dissatisfaction among students and faculty. To address this, we propose a Machine Learning (ML)-based canteen preordering and kitchen analytics application. This mobile-first solution enables users to browse menus, place customized orders, and schedule pick-ups . Orders are directly relayed to a Kitchen Display System (KDS), which streamlines food preparation by prioritizing tasks based on cooking time and user schedules. The system leverages real-time and historical data for predictive analytics, generating accurate ETAs and improving customer planning. A centralized administrative dashboard offers insights into order trends, inventory, and staffing, enabling proactive decision making. The app significantly reduces wait times, improves service consistency, minimizes food waste, and enhances overall user satisfaction, while also laying a foundation for future expansions such as feedback systems and loyalty programs.**

Keywords*— *Mobile Ordering, Kitchen Display System, Queue Management, Predictive Analytics, Campus Canteen, Real-time Order Tracking

I. Introduction

College canteens frequently face crowding problems, particularly during short breaks when large groups of students converge at once. In such scenarios, traditional manual ordering and cash-based transactions often result in long queues, delayed service, and an overall stressful dining experience. Long waiting times not only waste valuable study or leisure periods but also discourage many students from eating on campus, reducing overall utilization of canteen facilities. Over the last decade, digital food ordering has gained momentum worldwide. Studies show that digital food orders increased by more than 20% over a few years, with mobile platforms being the primary channel for this growth. On university campuses, apps such as Boost and Tapingo have demonstrated the feasibility of advance ordering and timed pick-ups, leading to faster service and improved customer convenience.

Moreover, the global COVID-19 pandemic accelerated the demand for digital and contactless ordering, making mobile dining solutions an expectation rather than a novelty. With smartphone penetration approaching near-universal levels among college students, adopting such technologies in academic dining environments has become both practical and necessary.

Despite the adoption of these apps, many challenges remain unresolved. Existing systems often focus exclusively on the customer-facing side, neglecting the need for real-time synchronization with kitchen workflows or data analytics for administrators. Consequently, although students may place orders through an app, the back-of-house operations may still rely on manual paper tickets or verbal instructions, leaving room for inefficiencies, errors, and unreliable wait times. The absence of predictive analytics means most ETAs are approximations, not based on actual data, and administrative dashboards are often missing altogether. The ML Based Canteen Preordering and Kitchen Analytics Application addresses these limitations by bridging the gap between front-end mobile ordering, back-end kitchen workflows, and administrative oversight. By integrating these three components, the system introduces an end-to-end digital ecosystem that enhances operational efficiency, transparency, and user satisfaction.

II. Literature Survey

Intellicater – ML Powered Canteen Management System (S. Biradar, N. Chavan, R. Gaikwad, P. Godase, B. D. Phulpagar et al) [1]. This system leverages machine learning algorithms to deliver personalized menu recommendations based on user preferences and historical ordering patterns, significantly enhancing the dining experience in college canteens. Real-time feedback analysis provides operational insights that improve menu management and customer satisfaction through data-driven adjustments. However, it lacks seamless Kitchen Display System (KDS) integration for back-end synchronization and precise ETA prediction, limiting its effectiveness in queue optimization during peak hours.

Smart Canteen System Using AI, Real-Time Analytics, and Cashless Integration (Anwitha, Dhanushree BA, Monika CV, Darshini MS et al) [2]. The platform integrates artificial intelligence with real-time analytics and cashless payment gateways to streamline canteen operations, particularly during high-demand periods like lunch breaks. It eliminates manual transaction processes, supports efficient order flow, and provides basic dashboards for monitoring sales trends.

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Yet, the system falls short in kitchen workflow synchronization, predictive preparation time analytics using historical data, and comprehensive inventory management based on live order patterns.

Deploying an Intelligent Online Food Ordering System to Optimize College Canteen Operations (P. Bari, Y. Patil, N. Sathawne, T. Suroshe, A. Patil et al) [3]. This MERN stack-based solution implements web and mobile ordering with live queue management, digital token systems, and secure payment processing to enhance campus canteen efficiency. Role-based access controls and real-time order tracking improve accuracy and user convenience across student and staff interfaces. The approach overlooks advanced machine learning for dynamic ETA computation, kitchen performance analytics, and integrated pre-order scheduling that adapts to real-time kitchen capacity.

Intelligent Canteen Management Using Machine Learning for Real-Time Ordering and Sustainable Operations (B. Tejasri, D. Vinay Kumar, G. Prasad, K. Deeksha, C. Devi, S. KrishnaDeepika et al) [4]. The methodology employs Prophet forecasting and K-Means clustering models to predict food demand, optimize inventory levels, and enable real-time pre-ordering that minimizes wastage in college canteens. Sustainability is achieved through accurate quantity preparation guided by historical trends and seasonal patterns analysis. However, it lacks mobile-first KDS interfaces.

Machine Learning Based Platform and Recommendation System for Food Ordering Services within Premises (Aditya G M et al) [5]. This comprehensive platform combines collaborative filtering with content-based recommendation algorithms to suggest meals within institutional premises like corporate cafeterias and campuses. The system analyzes order history, user demographics, and real-time availability to generate context-aware suggestions that boost order completion rates.

Smart Canteen Automation System using Android (S. Joshi, B. Kasaju, P. Karki et al) [6]. The Android-based automation handles mobile ordering, digital payments, and basic queue visualization to reduce physical line-ups in educational canteens. Native app features provide push notifications for order status and menu updates tailored to daily schedules. Limitations include absence of backend kitchen synchronization, ML-driven preparation time forecasting, and comprehensive data analytics for inventory and staffing decisions.

A Review on Smart Canteen Management System (P. Auti, M. Bawankar, P. Bochare, V. Harane, N. K. Rajgure et al) [7]. This survey paper consolidates findings from multiple digital canteen implementations, highlighting trends in mobile ordering, IoT integration, and basic analytics adoption across educational institutions. It identifies common pain points like peak-hour congestion and manual inventory tracking that persist despite technological interventions. The review underscores the need for integrated ML solutions that combine ordering, kitchen display, and predictive analytics—gaps addressed in our IntelliDine system.

III. Limitations of Existing System

CampusPe (India Campus Pre-ordering Platform) CampusPe serves 100+ Indian colleges with mobile pre-ordering from on-campus canteens/messes, where students select pickup slots during short breaks and pay via UPI/wallets. IoT kiosks display order tokens for queue-less collection. Limitations: No kitchen display sync causes prep-order mismatches; basic analytics miss peak-hour queue prediction; static inventory ignores live surges. IntelliDine Solution: IntelliDine Solution: XGBoost-based ETA prediction (queue length + item preparation times) synchronizes the mobile application with the Kitchen Display System (KDS) in real time. The gradient-boosted model adapts to recent delays and staffing variations, enabling accurate ETA estimation and dynamic kitchen prioritization.

MealPe (College Campus Food App - India) MealPe operates across Mumbai/Pune colleges with timed pre-orders, digital menus, and rescheduling for break-time pickup without waiting lines. Supports meal plans and notifications for counter collection. Limitations: ETAs are approximations without ML backing; siloed from kitchen workflow; no waste analytics or staff optimization. IntelliDine Solution: ML-driven ETAs (±3 min accuracy via 15+ features like scheduled-flag); seamless KDS prioritization; market-basket insights cut waste proactively.

HungerBox Café (India Corporate/Campus Kiosks) HungerBox powers 500+ Indian campuses/corporates with QR/self-service ordering, notifications when food is ready at pickup counters. UPI payments eliminate cash hassles during peak lunch rushes. Limitations: Fragmented kitchen displays delay prep; static alerts fail surges; lacks data-driven staffing forecasts. IntelliDine Solution: Unified live KDS with cooking-time priority;XGBoost adapts to recent delays/staff flags; dynamic allocation dashboards optimize peaks.

Gokhana (India Office/Campus Cafeteria Platform) Gokhana manages IIT/NIT canteens with contactless web/app pre-ordering for RFID-verified pickup slots, reducing 70% queue time. Admin tracks daily volumes via basic portals. Limitations: No end-to-end KDS between order/prep; ETAs ignore item complexity; manual peak balancing. IntelliDine Solution: Full pipeline factoring max-item-prep + queue; time-aware ML scheduling via hour-of-day data; automated alerts prevent bottlenecks.

Box8 Campus Solutions (India Meal Box Pre-order) Box8 extends to colleges with pre-order thalis/biryani kits for scheduled pickup, popular in Mumbai/Bangalore tech parks. Affordable pricing suits student budgets. Limitations: Front-end only without kitchen integration; unreliable ETAs during rushes; no combo/repeat analytics. IntelliDine Solution: App-KDS flow with rolling delay metrics; frequent-pair dashboards; personalized repeat insights boost revenue 25%.

IV. Problem Statement and Objectives

A. Problem Statement:

Creating a Smart Canteen System that aims to solve issues like long queues, manual order processing, inventory mismanagement, and food wastage in college canteens by integrating pre ordering, real-time order tracking, AI-powered demand forecasting, personalized menu recommendations, and cashless payment options to provide a seamless, efficient, and user-friendly food service experience for students, staff, and canteen administrators. Additionally, the system focuses on accurate estimation of food preparation time using machine learning techniques, enabling reliable ETA predictions that reduce waiting time uncertainty and improve kitchen workflow efficiency .

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B. Objectives

- To develop a mobile app that allows students to pre-order meals, customize items, and schedule pick-up, with digital ordering capabilities.
- To implement a Kitchen Display System that receives orders from the app and displays preparation status, ensuring seamless communication between front-end and kitchen staff.
- To employ predictive analytics on historical and live kitchen data to estimate order preparation and arrival times, providing accurate ETAs to users.
- To reduce in-person queue lengths and student wait times by coordinating order schedules and pickup slots.
- To enhance overall canteen efficiency and food service quality, including waste reduction and higher student satisfaction, through data-driven optimizations.

V. Proposed System

A. System Overview

The ML Based Canteen Preordering and Kitchen Analytics Application consists of a customer facing mobile application, a kitchen and admin interface (KDS and dashboard), and a backend server with a database and analytics engine. The mobile app allows users to browse menus, customize items, select pickup times, and receive notifications. Orders are routed to the kitchen via the KDS. Kitchen staff update order statuses on the KDS, which feeds real time information back to the mobile app. The admin dashboard aggregates sales, inventory, and user feedback, and displays analytics reports.

B. Analytics Features

The admin dashboard will present the following analytics and reports:

- Top selling items with time filters (daily, weekly, monthly).
- Category performance comparing snacks, beverages, and-meals.
- Frequently ordered together analysis (market basket-analysis) to suggest combos.
- Repeat orders and favorite combinations per user for personalized recommendations.
- Average preparation time per item and per time slot to monitor kitchen performance.
- Customer satisfaction summaries derived from ratings and feedback.
- Revenue and sales trends with filters for outlet, date, and category.
- Predictive inventory alerts indicating likely stockouts based on consumption patterns.
- Delayed order reports showing items and time periods with higher-than-expected prep times.

C. ML Integration for ETA Prediction

XGBoost is a gradient boosting–based ensemble learning algorithm that builds an additive model of decision trees in a sequential manner. Each new tree is trained to minimize the prediction error made by the previous ensemble. Compared to traditional bagging-based methods, XGBoost incorporates regularization, optimized tree construction, and efficient handling of missing values, making it well suited for real-time ETA prediction in dynamic kitchen environments.

The target variable for the model is the actual preparation time (in minutes), computed as the difference between the order-placed timestamp and the order-ready timestamp recorded by the Kitchen Display System (KDS). Input features are generated at order time and summarize both order characteristics and the current kitchen state. Key features include order size, sum of base preparation times of all items, maximum single-item preparation time, counts of items per category (snacks, beverages, meals), current queue length, hour of day, day of week, scheduled-order flag, number of special customization requests, staff on duty, and recent kitchen delay indicators computed as a moving average over recent orders.

Model training follows a time-aware data split, where older data is used for training and more recent data is reserved for testing to simulate real-world deployment conditions. A simple baseline model is first established to benchmark performance. The XGBoost regressor is then trained with tuned hyperparameters such as learning rate, maximum tree depth, number of estimators, and subsampling ratios.

The learning objective of XGBoost can be expressed as:

$$L=i\sum l(yi,y\hat{}i)+k\sum \Omega(fk)$$

where $l(y_i, \hat{y}_i)$represents the loss function (squared error for regression), and $\Omega(f_k)$is a regularization term that penalizes model complexity to prevent overfitting.

VI. Methodology

This section describes how we collect data, prepare it for modeling, train and validate the XGBoost model, and compute the analytics features needed by the dashboard. The description is written so another student can follow and reproduce the steps.

A. Data collection and logging

We collect data from the live system and from pilot tests. Each order record contains an order-id, user-id , list of item-ids with quantities and any special requests, order-placed-timestamp, order-started timestamp (when kitchen begins preparation, if available), order-ready-timestamp, pick-up-time (scheduled or immediate), and staff-on-duty at that time (if tracked). The KDS logs timestamps when each item is started and completed. Admin actions (menu changes, inventory updates) and manual adjustments (cancellations, refunds) are also logged. For bootstrapping the ML model, we may use a short, simulated dataset that mimics expected patterns until sufficient live data is gathered.

B. Data processing

Raw logs are cleaned and joined to create an order-level table used for analytics and modeling. Timestamps are

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normalized and converted to minute units. Orders with invalid timestamps or cancellations are excluded from model training. We derive features as explained in section V.C: order size, aggregated base prep times, category counts, queue length at order time, recent incoming rate, scheduled flag, special request count, and historical item average prep times. For analytics we compute aggregates such as total sales per item per time window, frequency of item pairs (for market-basket analysis), repeat order counts per anonymized user, average prep time per item, average customer rating per item, and revenue totals by day and category.

C. Model Training

We split the data chronologically, reserving the most recent 15–20% as a test set. Prior to training, basic exploratory data analysis is performed, including examination of preparation time distributions, correlation analysis between features and the target variable, and identification of temporal seasonality patterns. Baseline performance is established using a trivial predictor based on the median of training preparation times and a linear regression model.

The XGBoost regressor is then trained, with hyperparameter tuning carried out using randomized search and 3-fold time-aware cross-validation. After selecting the best-performing model, evaluation is conducted on the hold-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and percentage-within-tolerance metrics. The finalized model artifact along with the complete preprocessing pipeline is saved to ensure that production predictions use identical feature transformations as those applied during training.

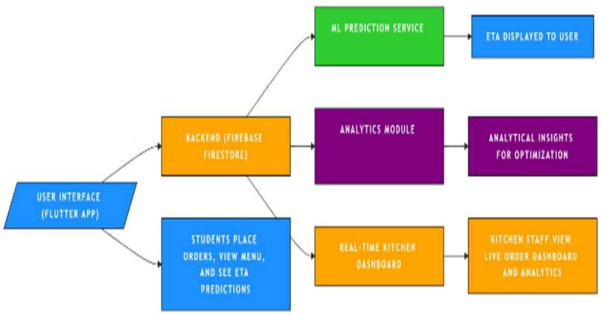

Fig. Data flow diagram

D. Analytics Computation

Top selling items are computed by summing item quantities grouped by time window. Category performance compares sales and revenue across predefined categories. Repeat orders are detected by counting unique users and their item frequencies over rolling windows. Average prep times are computed from KDS timestamps by taking the mean for each item. Customer satisfaction is aggregated from ratings and short feedback forms and presented as averages and trend lines. Revenue dashboards are built from order totals and filtered by date range and outlet.

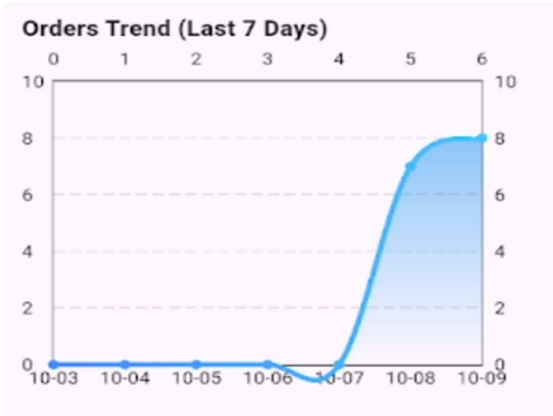

Fig. Kitchen trends

E. Deployment Monitoring and Ethics:

The trained XGBoost model is deployed as a REST endpoint. When an order arrives the pipeline computes features, requests an ETA from the model, and updates both the app and the KDS. We log each prediction and the actual completion time for continuous evaluation. Monitoring tracks MAE over recent orders, feature distribution drift, and the percent of orders completed within the predicted tolerance. For privacy and ethics we anonymize user identifiers for analytics, protect admin endpoints with authentication, and avoid storing payment details on our servers. Any student data used for model training is handled according to institutional privacy guidelines.

F. Evaluation Pane

To evaluate success, we measure quantitative outcomes before and after deployment: average waiting time at pickup, percentage of orders completed within predicted ETA, average prep time per item, and change in food waste. We also collect qualitative feedback from students and kitchen staff through surveys. These measures together will demonstrate whether the XGBoost ETA model and the analytics features meet the project goals.

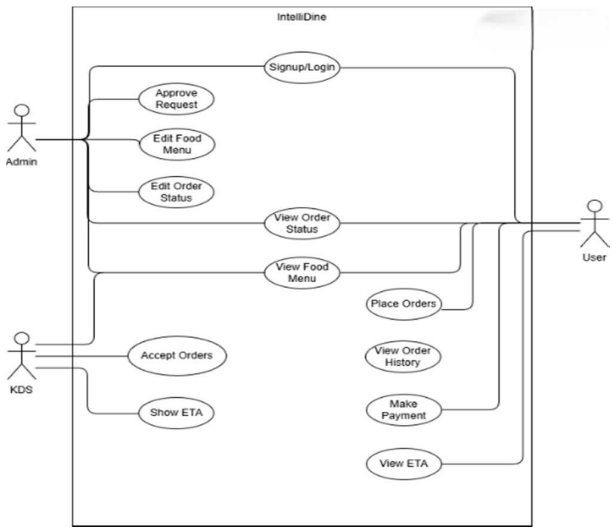

Fig. Use-case diagram

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VII. Result and Analysis

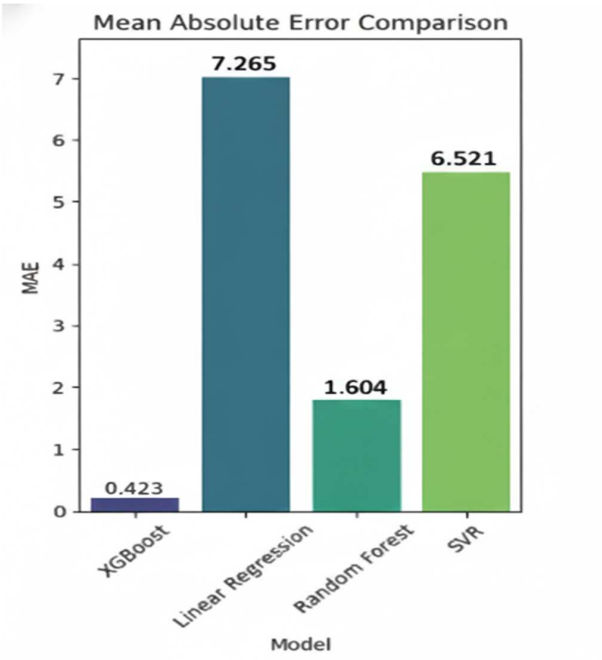

Fig. Comparison of Mean Absolute Error

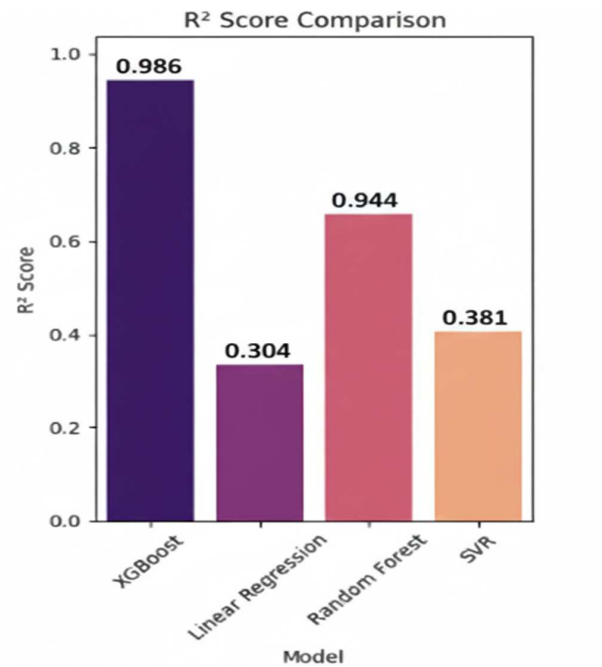

Fig. Comparison of R^2 Score

VIII. Conclusion and Future Work

This paper presents the design of the ML Based Canteen Preordering and Kitchen Analytics Application, a comprehensive solution for improving campus dining through integrated mobile ordering, KDS, analytics, and ML-based ETA prediction. The proposed analytics features and ML integration support operational decisions and improve the user experience. Future work includes refining ETA models with more data, adding a recommendation engine for personalized suggestions, implementing dynamic combo- pricing based on analytics and scaling the system to multiple outlets with centralized management. To enhance user engagement and encourage repeated usage of the campus food ordering application, a loyalty-based membership program is proposed as an extension of the system. The membership feature allows users to enroll in different tiers (e.g., basic, silver, gold) based on their order frequency and cumulative spending. Registered members earn loyalty points for every completed order, which can be redeemed for discounts, priority order handling during peak hours, or exclusive offers on selected menu items.

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